■ Detection Accuracy Enhancement Using

YOLO26m + SAHI

Manufacturing Surveillance Person Detection System

■ Objective

The goal of this task was to **improve person detection accuracy and reliability** in manufacturing surveillance streams.

As part of continuous system optimization, we evaluated detection performance and upgraded our pipeline from **YOLOv8 (YOLOv8l)** to **YOLO26 (YOLO26m) + SAHI**.

This migration resulted in **higher recall, better small-object detection, and more stable real-time inference**.

■ Model Evolution Overview

■ Earlier Approach — YOLOv8l

Observed Behavior:

- Good general detection performance
- Accurate for medium/large persons
- Real-time capable

Gaps noticed during evaluation:

- Reduced detection of **far-distance/tiny persons**
- Missed **partially occluded workers**
- Lower recall in **crowded scenes**
- Detail loss when processing high-resolution frames

■ Upgraded Approach — YOLO26m + SAHI

Outcome After Upgrade:

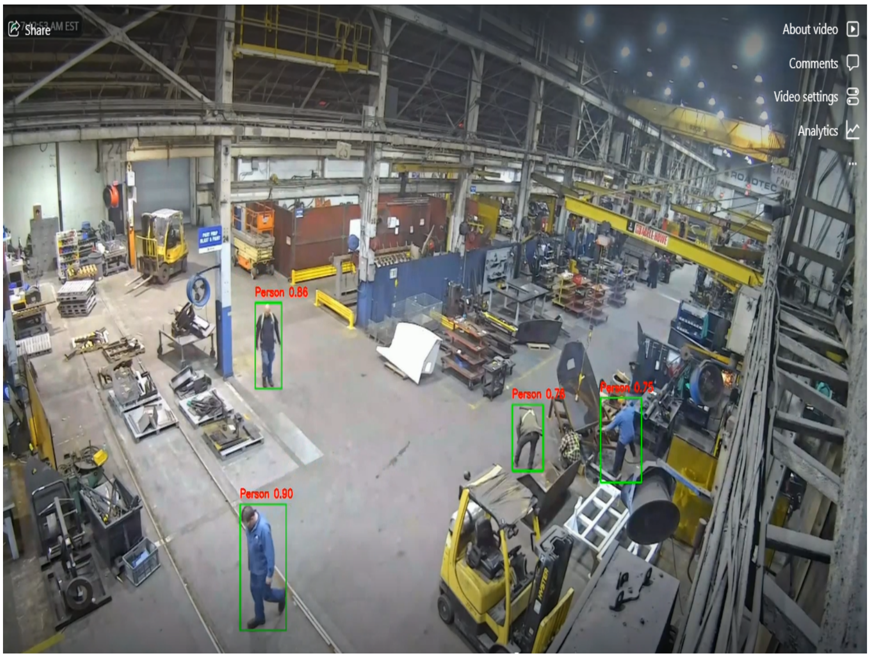
- More persons detected at long distances
- Better recognition of tiny targets
- Stable detection in dense environments
- Higher overall recall
- Consistent real-time inference

■ Visual Performance Comparison

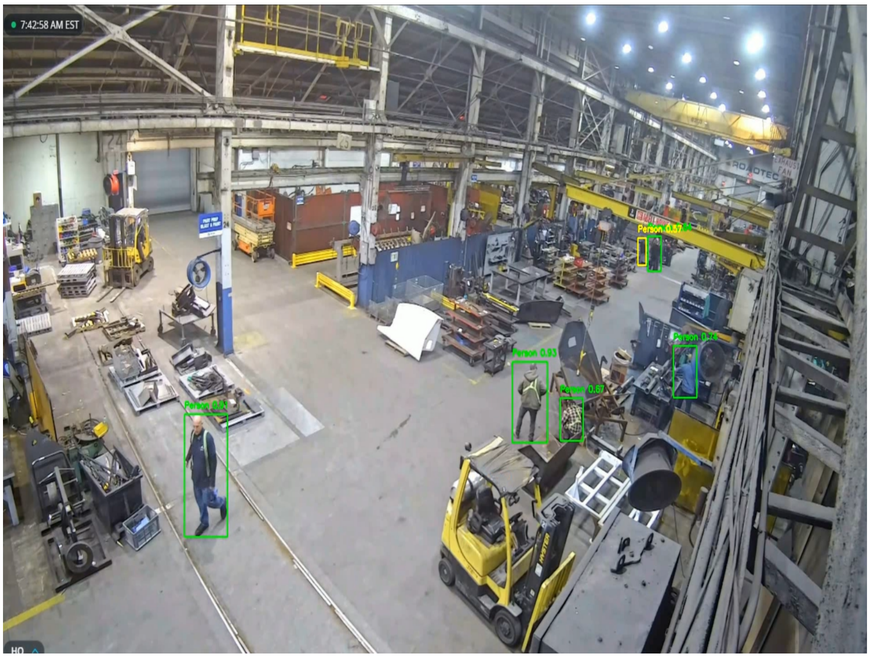
Comparison 1: GEN2 Dataset

The following images demonstrate the detection improvement on the GEN2 surveillance dataset. Notice the enhanced detection of distant and small objects with YOLO26m + SAHI.

YOLOv8l Detection:



YOLO26m + SAHI Detection:



Comparison 2: HP Dataset

This comparison showcases performance on the HP surveillance dataset. The YOLO26m + SAHI combination provides significantly better coverage and accuracy.

YOLOv8l Detection:

YOLO26m + SAHI Detection:

■ YOLO26m — Technical Capabilities Delivered

1. Small-Object Focused Learning

Improved label assignment and training strategy increases sensitivity to **tiny and distant persons**.

2. End-to-End Detection (No NMS Dependency)

Direct predictions reduce box suppression and overlapping miss detections. Result → **Better crowded-scene reliability**

3. Faster Inference Pipeline

Optimized architecture enables lower latency, higher FPS, and smooth CCTV processing.

4. Robust Feature Extraction

Enhanced backbone improves occlusion handling, complex background separation, and machinery/person differentiation.

5. Efficient Compute Usage

Medium variant provides near-large accuracy with lower GPU/CPU consumption. Ideal for edge/industrial deployments.

✂■ SAHI — Inference Strategy Enhancements

1. Image Slicing Mechanism

Large frames are split into smaller tiles before detection.

2. Enhanced Pixel Visibility

Small persons appear larger inside each tile → easier detection.

3. Higher Recall Without Retraining

Improves detection using the same model weights.

4. Scalable to High Resolution (1080p/4K)

Maintains detail without downscaling.

5. Plug-and-Play Integration

Works directly with YOLO detectors without architectural changes.

■■ How the Combined System Works

```
Detection Pipeline:
Video Stream
    ↓
Frame Extraction
    ↓
SAHI → Slice into tiles
    ↓
YOLO26m inference per tile
    ↓
Box fusion & merge
    ↓
Final detections
    ↓
Monitoring / Alerts / Analytics
```

Performance Comparison

Metric	YOLOv8l	YOLO26m + SAHI
Small person detection	Medium	High
Far-distance recall	Low–Medium	High
Occlusion handling	Moderate	Strong
Crowded scenes	Partial misses	Stable
High-res support	Downscale needed	Native
FPS	Good	Better
Overall reliability	Good	Excellent

Final Outcome

Achievements:

- Increased person detection recall
- Reduced missed detections
- Improved small & distant object recognition
- Stable performance in dense industrial scenes
- Real-time edge deployment ready

Summary Statement

As part of our detection optimization initiative, we migrated from YOLOv8l to YOLO26m integrated with SAHI sliced inference. This upgrade significantly improved small and distant person detection, enhanced robustness in crowded and occluded manufacturing environments, and delivered higher real-time accuracy without increasing computational cost.